

# The Ontological Boundary of Quantum Probability

## *Pre-Collapse Possibility versus Realised Physical Fact*

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### **Abstract**

This paper identifies a fundamental ontological boundary in quantum mechanics: the collapse event. The pre-collapse domain is governed by the Schrödinger equation and describes the complete probability amplitude structure of unrealised physical possibilities. The post-collapse domain consists of definite physical events subject to the definite-state descriptions provided by classical and relativistic physics. The failure to draw this boundary explicitly has generated compounding confusion in quantum foundations, including the conflation of mathematical probability structures with physical distributions of matter and energy, and the misapplication of the Heisenberg uncertainty principle beyond its proper descriptive domain. This paper argues that Heisenberg's uncertainty principle is a law of the pre-collapse domain whose subject matter does not appear in physical reality as indeterminacy. The long-standing debate between Einstein and Bohr is reframed rather than adjudicated: both positions are shown to be coherent within their respective domains, the apparent contradiction arising from the absence of an explicit boundary between them. The Many Worlds interpretation is examined as a consequence of extending the ontology of the pre-collapse domain beyond its proper descriptive scope. The central difficulty in constructing a quantum theory of gravity is reconsidered as a possible category error in treating probability amplitudes as direct sources for classical spacetime curvature. Drawing the collapse boundary explicitly reorients the foundational questions of quantum mechanics and identifies which longstanding problems dissolve, which are reformulated, and which remain genuinely open.

**Keywords:** quantum collapse; ontological boundary; Heisenberg uncertainty principle; wave function; pre-collapse domain; post-collapse domain; measurement problem; Many Worlds interpretation; quantum gravity; Bohr-Einstein debate; decoherence; probability amplitude; physical reality; descriptive domain; category error

# Chapter One: The Boundary That Was Never Drawn

## 1.1 The Question Lorentz Asked

On 24 October 1927, the Fifth Solvay Conference opened in Brussels. Every significant architect of the new quantum theory was present: Einstein, Bohr, Heisenberg, Planck, Curie, de Broglie, Dirac, Pauli, Schrödinger. The president of the meeting, Hendrik Lorentz, opened the general discussion with a question that cut directly to the matter at hand: "Could one not maintain determinism by making it an article of faith? Must one necessarily elevate indeterminism to a principle? [1]"

Lorentz's question was not a technical objection. It was a philosophical challenge, and it identified with precision the fault line along which the entire subsequent debate would run. The assembled physicists had built a mathematical framework of extraordinary predictive power. What they had not settled—and what Lorentz was asking them to confront—was a prior question: what is this framework actually describing? Is quantum mechanics a theory of physical reality as it exists, or a theory of possibilities as they are calculated?

The debate that followed between Einstein and Bohr lasted, in one form or another, for the rest of both men's lives. It has been characterised as a dispute about determinism, about realism, about the completeness of quantum theory, about whether God plays dice. All of these framings contain partial truth. But underneath every one of them, the same unresolved question recurs: does quantum mechanics describe a physical world that exists, or does it describe a mathematical structure of possibilities that has not yet become a physical world?

This question was never formally answered at the Fifth Solvay Conference. It was never formally answered at the Sixth. It was not answered by the EPR paper of 1935, nor by Bohr's response to it, nor by Bell's theorem in 1964, nor by the experimental tests of Bell inequalities in the decades that followed. It remains, in formal terms, unanswered today—because the physics community treated it as a question about interpretation, a matter of philosophical preference, rather than a question with a structural answer derivable from the theory itself.

This paper argues that the question does have a structural answer, and that the answer was already implicit in the formalism by 1927. The failure to identify it produced a century of compounding confusion: the conflation of mathematical possibility with physical existence, the demand that a theory of realised physical facts accommodate a description domain that belongs entirely to unrealised probability, and a proliferation of interpretive frameworks—many of them spectacular in their mathematical elaboration—that rest on a single foundational category error.

The answer begins with a boundary. And the boundary begins with collapse.

## 1.2 Two Domains, One Formalism

Quantum mechanics operates across two structurally distinct domains, and it has always done so. The failure to recognise them as distinct—to treat them as continuous rather than separated by a hard boundary—is the source of the confusion this paper addresses.

The first domain is the pre-collapse quantum state. Before any measurement interaction terminates the evolution of a quantum system, that system is described by a wave function. The wave function assigns probability amplitudes to all possible outcomes of any given measurement. It evolves deterministically according to the Schrödinger equation. It is, in every formal sense, a complete and precise mathematical object. It is not vague, not approximate, not a placeholder for missing information. It is an exact description of a probability structure.

The critical point—the one that has been consistently underemphasised in the century of interpretation debates—is that this probability structure is not a physical fact. It is a mathematical fact. The wave function assigns probability amplitudes to possible outcomes, from which the relative likelihoods of different results are derived. It does not tell us this because nature is hiding something, or because our instruments are imprecise, or because we have not looked carefully enough. It tells us probabilities because probabilities are all there is to tell, at this stage, about a system that has not yet produced a definite physical result. The electron is not secretly in one position while the wave function describes many positions. The electron's physical situation, prior to collapse, is exactly and fully captured by the quantum state and its associated probability amplitudes. There is no additional fact about position that the wave function omits.

This is the quantum description domain: the domain of unrealised physical possibility, described with mathematical exactness.

The second domain is the post-collapse physical event. When a quantum system interacts with a measurement apparatus—or, more precisely, when the quantum evolution produces a definite macroscopic outcome through the process known as collapse—a physical fact comes into existence. The electron is detected at position A. The photon passes through the left slit. The radioactive atom decays at time T. These are not probability distributions. They are events. They exist in physical reality in the way that all physical events exist: as singular, located, irreversible occurrences in spacetime.

These two domains are not points on a continuum. They are not related by a smooth transition. They are separated by the collapse event, which is a boundary in the most fundamental sense: the moment at which mathematical possibility becomes physical fact, at which the description domain of quantum mechanics gives way to the event domain of classical reality.

The wave function does not gradually become a physical event. The probability distribution does not slowly crystallise into an outcome. Collapse is a transition, not a process—and what lies on either side of it belongs to categorically different ontological territory.

This boundary was implicit in the quantum formalism from the moment the formalism existed. The Schrödinger equation governs evolution within the first domain. The Born rule governs the transition between domains. Classical mechanics governs behaviour within the second domain. Three distinct mathematical frameworks, three distinct regimes—and a single hard boundary between the first and the third [2].

What the physics community failed to do in 1927, and has not fully done since, is to state this boundary explicitly and draw the consequences from it.

### 1.3 Heisenberg's Principle and Its Hidden Boundary Condition

Werner Heisenberg published his uncertainty principle in March 1927 [3,4], eight months before the Fifth Solvay Conference. The principle states that the position uncertainty and momentum uncertainty of a quantum particle cannot both be made arbitrarily small simultaneously. In its canonical form:

$$\Delta x \cdot \Delta p \geq \hbar/2$$

This relationship is not a statement about measurement precision. It is not a consequence of instruments disturbing particles. It is a structural feature of quantum states: a wave function that is

sharply localised in position space necessarily contains a broad distribution of momentum components, and vice versa. This follows from the mathematics of Fourier decomposition—a narrow spatial wave packet requires the superposition of many wavelengths, and wavelength corresponds to momentum. The uncertainty is not imposed on the quantum state from outside; it is constitutive of what quantum states are.

In 1927, the uncertainty principle was formulated and understood entirely within the quantum description domain—the domain of wave functions, probability amplitudes, and unrealised physical possibilities. At the time of its formulation, the concept of wave function collapse had not yet been developed into a formal theoretical element. Collapse as an explicit feature of quantum measurement theory—the idea that measurement produces a definite outcome by projecting the wave function onto one of its eigenstates—emerged through the discussions at and after the Fifth Solvay Conference, and was consolidated in John von Neumann's mathematical treatment of quantum mechanics, published in 1932 [5].

This chronology matters more than it has been recognised to matter. Heisenberg's principle was formulated, stated, and understood as a property of quantum states. Its entire content concerns the internal structure of wave functions. Its domain of application is, by construction, the pre-collapse quantum description domain. At the moment of its formulation, there was no other domain to consider—collapse had not yet been formally integrated into the theory.

When collapse was subsequently introduced, the boundary between the two domains became explicit. The question that should have been asked—and was not asked with sufficient precision—is this: does the uncertainty principle apply within the post-collapse domain?

The answer, examined carefully, is that it does not—not to the specific event that collapse has produced. When a quantum system undergoes collapse and a definite physical outcome occurs, that outcome is a fact. The electron is at position A. Not probably at A, not approximately at A, not at A with some residual spread that Heisenberg's principle requires—at A, as a definite physical event. The wave function that described the probability distribution of possible positions prior to collapse no longer exists as the description of this system's current situation. A new quantum state may be assigned following the measurement, and that new state will itself be subject to the uncertainty principle. But the uncertainty principle governing the new state is a statement about what further measurements on the system would yield—not a statement that the completed measurement result is somehow still uncertain.

To say that "Heisenberg's principle remains valid after collapse" is to conflate the principle's mathematical persistence with its physical applicability. The mathematical relationship  $\Delta x \cdot \Delta p \geq \hbar/2$  is always true of any quantum state. But the pre-collapse quantum state that described this particular event—the state whose position-momentum uncertainty was the subject of Heisenberg's analysis—no longer exists as the description of a physical situation. Its subject matter has been resolved into a definite outcome. The principle has not been violated; its object of application has ceased to exist.

Consider the analogy of a probability distribution for tomorrow's weather. Before tomorrow arrives, a meteorologist can assign probabilities: 70% chance of rain, 30% chance of sun. This probability distribution is a precise mathematical object. It correctly describes the epistemic and physical situation before the weather event occurs. When tomorrow arrives and it rains, the probability distribution does not continue to apply to yesterday's weather. It is not the case that yesterday's rain is "still" 70% probable. The event has occurred. The probability description was valid until the event; after the event, the description's object no longer exists as an open possibility—it has become a closed fact.

Heisenberg's uncertainty principle stands in exactly this relationship to the collapse event. It governs the quantum description domain with complete validity. The moment collapse occurs and a definite physical outcome exists, the domain it governs—the domain of unrealised quantum possibilities for that event—has closed. The principle's applicability to that specific event terminates at the collapse boundary, not because the principle is wrong, but because its subject matter is gone.

This is not a minor technical clarification. It is a structural reorientation of how the uncertainty principle is understood. Heisenberg's principle is a law of the quantum description domain. It is not a law of physical events. The two have been conflated for a century because the boundary between the domains was never explicitly drawn—and the consequences of that conflation extend far beyond the uncertainty principle itself, into the deepest problems of quantum theory and its relationship to the rest of physics.

The boundary exists. It runs through collapse. Everything that follows in this paper depends on recognising it for what it is.

## **Chapter Two: One Hundred Years of Ontological Drift**

### **2.1 The Fifth Solvay Conference and the Question That Was Left Open**

The Fifth Solvay Conference of October 1927 is remembered, in most popular accounts, as the moment when Einstein began his long argument with Bohr about the completeness of quantum mechanics. This framing is accurate as far as it goes, but it obscures something more fundamental that happened in Brussels that week—or rather, something that failed to happen. The physicists assembled in the Hôtel Métropole did not merely disagree about the interpretation of quantum mechanics. They failed, collectively, to identify the structural question that their disagreement was actually about. As a result, the debate they initiated was conducted across a conceptual gap that no one had explicitly mapped, and the confusion that gap produced has compounded with every decade since.

To understand what went wrong in 1927, it is necessary to reconstruct not just what was said, but what was assumed—and what those assumptions concealed.

By October 1927, the mathematical machinery of quantum mechanics existed in two equivalent formulations. Heisenberg's matrix mechanics, developed in 1925, represented quantum observables as matrices whose non-commutativity encoded the uncertainty relations directly into the algebraic structure of the theory. Schrödinger's wave mechanics, developed in 1926 [6], represented quantum states as wave functions evolving in configuration space according to the equation that bears his name. Both formulations gave identical predictions. Both described, with complete mathematical precision, the probability structure of quantum systems prior to measurement [7].

What neither formulation addressed—what both, in a sense, deferred—was the question of what happens when a measurement actually occurs. Schrödinger's equation is linear and deterministic: a wave function evolves smoothly and continuously, and if that evolution is all there is, then measurement produces no definite outcome—it produces a superposition of all possible outcomes, each weighted by its probability amplitude. The Born rule, introduced by Max Born in 1926, provided a prescription for calculating measurement probabilities from wave function amplitudes, but it did not explain how or why a single definite outcome is obtained from a superposition. The gap between the deterministic evolution of the wave function and the single definite outcome of a measurement was already present in the formalism. It was not noticed as a gap because the formalism worked—the predictions were confirmed, the mathematics was tractable, and the physical consequences were revolutionary.

At the Fifth Solvay Conference, this gap was present in every discussion, but it was approached from the wrong direction. Einstein's objections were framed as objections to the completeness of quantum mechanics: if the wave function does not tell us where the electron actually is, only where it might be found, then the wave function is not a complete description of physical reality. Bohr's responses were framed as defences of the formalism's completeness: the wave function is the complete description, and the demand for a more precise specification of the electron's position prior to measurement reflects a misapplication of classical concepts to a domain where they do not apply. Both framings shared an assumption that neither man made explicit: that the question of what the wave function describes is the same question before and after a measurement occurs. This assumption is the source of the confusion.

Einstein's position, in its most direct form, was this: if quantum mechanics describes physical reality, then physical reality must have more definite structure than the wave function assigns to it. A theory that can only tell us probabilities is incomplete, because physical reality—the electron, the photon, the atom—has definite properties whether or not we measure them. His famous remark, made in various forms across many years, that the moon exists whether or not anyone looks at it, captures the core of this intuition. Reality is not constituted by observation. It is discovered by observation.

Bohr's position, in its most direct form, was this: the demand that quantum systems possess definite properties prior to measurement is a demand that belongs to classical physics, and quantum mechanics has demonstrated that classical physics does not apply at the microscopic scale. The wave function is not an incomplete description of a more precise classical reality. It is the complete description of a quantum reality, which is structured differently from classical reality. The uncertainty is not ignorance—it is a feature of nature itself.

Both positions contain genuine insight. Both positions, as stated, are also partially wrong—because both assume that the question of what quantum mechanics describes has a single answer that applies uniformly across the entire domain of the theory. The identification of the pre-collapse and post-collapse domains as structurally distinct, separated by a hard ontological boundary, dissolves the apparent contradiction between them. Einstein is right about post-collapse reality: once a physical event has occurred, it has definite properties, and those properties exist independently of whether anyone is observing them. Bohr is right about pre-collapse quantum states: the probability structure of an unrealised possibility is not an incomplete description of a hidden definite reality—it is the complete description of something that is genuinely not yet physically definite. The error shared by both men was to treat these as competing answers to a single question, rather than as correct answers to two different questions about two different domains.

This error was not corrected at the Fifth Solvay Conference. It was not corrected at the Sixth, in 1930, where Einstein introduced his famous thought experiment of a clock suspended from a balance inside a box, designed to challenge the energy-time uncertainty relation. It was not corrected in the EPR paper of 1935. It was not corrected by Bell's theorem in 1964. It remains uncorrected in the mainstream of quantum foundations research today [8,9,10].

The reason it was not corrected is straightforward: it was not identified. The assumption that the quantum formalism describes a single uniform ontological domain—that the pre-collapse wave function and the post-collapse physical event are both within the scope of the same descriptive framework—was never questioned. It was inherited as an implicit feature of the formalism and transmitted, unexamined, into every subsequent interpretation debate. The interpretations multiplied—Copenhagen, Many Worlds, Pilot Wave, Consistent Histories, Relational Quantum Mechanics, QBism—but they all addressed the same question framed in the same way, with the same hidden assumption intact.

Bohr himself never claimed that the wave function describes physical reality. His position was more cautious: the wave function is the complete account of what can be said about a quantum system, and asking what the system "really is" prior to measurement is a question that quantum mechanics neither answers nor needs to answer [11]. This epistemological restraint was precise and deliberate.

The conflation of the descriptive domain with physical reality did not originate with Bohr. It entered through two successive steps. Von Neumann's 1932 mathematical formulation placed both the Schrödinger evolution and the collapse process within a single unified framework [5], implicitly treating them as two processes within the same ontological domain rather than as the governing law of one domain and the boundary event separating it from another. Everett then took the logic of that framework to its conclusion [12]: if collapse is merely one process among others within quantum mechanics, it can be eliminated entirely, and every branch of the wave function becomes equally real. Bohr rejected Everett's interpretation outright when Everett visited Copenhagen in 1959, but refused to engage with the structural question of why the wave function could not be treated as a description of simultaneously existing physical realities—the conceptual gulf, as contemporaries noted, proved too wide for any meeting of minds. Léon Rosenfeld, Bohr's closest collaborator, described Everett as "undescritably stupid and could not understand the simplest things in quantum mechanics" [13]—a response that reveals the depth of the rejection but not its justification. What began as Bohr's careful refusal to ontologise the wave function ended, through these two steps, as the full ontologisation of every branch of the universal wave function. The probability structure that Bohr had treated as a descriptive tool became, in Everett's hands, a catalogue of simultaneously existing physical realities.

## 2.2 The Historical Accumulation: How the Conflation Deepened

Between 1927 and the present, the conflation of the quantum description domain with the physical event domain did not remain static. It deepened, elaborated itself, and generated a succession of theoretical structures that inherited its foundational confusion and amplified it into increasingly elaborate forms.

The first major amplification occurred with the development of the measurement problem as a formal theoretical concern. John von Neumann's 1932 treatment of quantum mechanics, *Mathematische Grundlagen der Quantenmechanik*, was the first rigorous mathematical formulation of quantum theory as a complete physical theory, and it introduced a distinction that, while formally useful, also deepened the conceptual confusion. Von Neumann distinguished between two types of quantum evolution: the continuous, deterministic, linear evolution governed by the Schrödinger equation (which he called Process 2), and the discontinuous, probabilistic transition that occurs at measurement (which he called Process 1). This distinction correctly identified the two regimes of quantum theory. But von Neumann's treatment of Process 1—wave function collapse—as an additional dynamical process, operating alongside and distinct from Process 2, implicitly placed both processes within the same ontological domain. Collapse, in von Neumann's formulation, is not a transition between two domains; it is an event within a single quantum domain [5].

This framing set the terms for every subsequent treatment of the measurement problem. If collapse is a process within the quantum domain, then it must be governed by quantum mechanics—but quantum mechanics, as specified by the Schrödinger equation, does not produce collapse. The measurement problem, as it has been studied for nearly a century, is precisely the problem of explaining how a linear, deterministic quantum evolution can produce the non-linear, probabilistic outcomes that measurement actually yields. Enormous theoretical resources have been devoted to this problem: decoherence theory, quantum Darwinism, spontaneous collapse models, and the many-worlds interpretation are all, in different ways, attempts to solve it [12,14,15,16].

All of these attempts share the assumption that the measurement problem requires a solution within quantum mechanics—that the transition from wave function to definite outcome must be explicable in terms of quantum mechanical processes. This assumption is the direct consequence of treating pre-collapse and post-collapse as belonging to the same ontological domain. If they belong to different domains, separated by a hard boundary, then there is no measurement problem in this sense. The

Schrödinger equation governs the pre-collapse domain. Definite physical outcomes exist in the post-collapse domain. The transition between them is the collapse event—the boundary itself. There is no requirement that the Schrödinger equation explain how its own domain terminates, any more than a map is required to explain what lies beyond its edges.

The second major amplification came with the development of quantum field theory and the subsequent attempt to formulate a quantum theory of gravity. Quantum field theory extended the quantum formalism to relativistic systems, representing particles as excitations of underlying fields and describing their interactions with extraordinary precision. The Standard Model of particle physics, built on quantum field theory, is the most precisely confirmed physical theory in the history of science. But quantum field theory did not resolve the foundational confusion about the two domains. It extended the quantum description domain into relativistic spacetime, and it continued to treat the wave functional of the quantum field as a description of a single uniform ontological domain without an explicit treatment of the collapse boundary [17].

When the attempt was made to extend quantum mechanics to gravitation—to construct a quantum theory of gravity—the foundational confusion produced its most dramatic consequences. General relativity, Einstein's theory of gravitation, describes the geometry of spacetime as determined by the distribution of matter and energy within it. The Einstein field equations relate the curvature of spacetime to the energy-momentum tensor—a description of the actual, definite physical distribution of matter and energy in spacetime. General relativity is, fundamentally, a theory of the post-collapse physical domain: it describes physical events, definite configurations of matter and energy, and the spacetime geometry they produce.

Quantum mechanics, by contrast, describes—in its pre-collapse regime—a probability distribution over possible configurations of matter and energy. When the attempt is made to combine these two theories, the foundational question is unavoidable: what is the source term in the Einstein field equations when the matter and energy distribution is quantum mechanical, and therefore described by a probability distribution over possible configurations rather than a single definite configuration?

This is the quantum gravity problem at its most basic level, and it has resisted solution for nearly a century. Multiple approaches have been attempted: canonical quantum gravity, loop quantum gravity, string theory, and many others. None has succeeded in producing a consistent, empirically confirmed quantum theory of gravity. The reason—or at least one fundamental reason—is that the project attempts to apply the Einstein field equations, which require a definite physical distribution of matter and energy, to a domain where matter and energy are described by a quantum probability distribution. This is a category error. General relativity cannot take a quantum probability distribution as its source term, because a probability distribution is not a physical distribution of matter and energy—it is a description of possible physical distributions, none of which has yet been physically realised as a fact [18,19,20].

To demand that spacetime geometry respond to quantum probability amplitudes is to demand that the physical universe respond to unrealised possibilities. It is to require that the post-collapse domain—the domain of definite physical facts, which is the domain general relativity governs—be determined by the pre-collapse domain—the domain of unrealised mathematical possibilities, which is the domain quantum mechanics governs before measurement. This demand is not a technical challenge to be overcome by more sophisticated mathematics. It is a structural impossibility arising from the conflation of two categorically distinct domains.

## **2.3 Einstein's Intuition: What He Was Right About and Why He Could Not Prove It**

Einstein's resistance to quantum mechanics has been characterised, in the standard historical account, as the stubbornness of a great physicist who could not accept the probabilistic structure of the new

theory—who insisted, against the evidence, that God does not play dice. This characterisation is both unkind and inaccurate. Einstein's objections to quantum mechanics were not emotional. They were structural, and they were directed, with remarkable precision, at the exact conceptual location where the theory's foundational confusion is deepest.

Einstein's core objection, stated in its most direct form, was this: quantum mechanics, as interpreted by Bohr and Heisenberg, does not describe physical reality as it exists independently of observation. It describes, at most, our knowledge of physical reality, or the results of our interactions with physical reality. A theory that cannot assign definite properties to physical systems between measurements is not a theory of nature—it is a theory of what we can say about nature given the limitations of our experimental access to it. As Einstein wrote to Schrödinger in 1935: physics describes reality. But we do not know what reality is; we only know it through the physical description [21,22].

This formulation is precise and important. Einstein is not claiming that quantum mechanics is wrong. He is claiming that it is incomplete—that it describes something real and important, but not the whole of what is physically real. The mathematical structure represented by the wave function is real as a component of the theory; the mathematical structure of quantum mechanics is accurate and confirmed. But the probability distribution is not itself the physical reality it is a distribution over. There is a physical reality—the electron in a definite location, the photon with a definite momentum—and the wave function describes our probabilistic relationship to that reality, not the reality itself.

In 1927, Einstein presented a series of thought experiments at the Solvay Conference designed to demonstrate that quantum mechanics, as interpreted by Bohr, led to inconsistencies or incompleteness. Each thought experiment was designed to show a situation in which definite information about a quantum system could be obtained without the disturbance that Heisenberg's uncertainty principle seemed to require. Bohr's responses to these thought experiments were, in each case, technically correct: Einstein had overlooked some physical constraint—the recoil of the measurement apparatus, the entanglement between the measured system and the measuring device—that preserved the uncertainty relations. Einstein's thought experiments did not succeed in demonstrating an inconsistency in quantum mechanics [1,7,23].

But the failure of each individual thought experiment should not be allowed to obscure the validity of the underlying objection. Einstein was not wrong to insist on the distinction between probability descriptions and physical reality. He was wrong about where the boundary lies. He assumed that the distinction between probability and reality runs through every quantum state—that every quantum system, at every moment, has definite properties that the wave function fails to capture. The correct distinction, as argued in this paper, is that the boundary runs through collapse: the pre-collapse quantum state is genuinely probabilistic, not a description of hidden definite properties; the post-collapse physical event is genuinely definite, not a quantum probability distribution that has been partially resolved. Einstein was right about the post-collapse domain—results are results, facts are facts, and the moon is where it is regardless of whether anyone measures it. Bohr was right about the pre-collapse domain—the wave function is the complete description of an unrealised possibility, not an incomplete description of a hidden definite reality. Neither man recognised that they were both right, about different domains.

The tragedy of the Bohr-Einstein debate is not that one man was right and the other wrong. The tragedy is that both men were right about half of the physical situation, and neither man had access to the conceptual framework that would have allowed them to see this. The framework requires identifying the collapse event as a hard ontological boundary—not a dynamical process, not a measurement interaction, not a decoherence phenomenon, but a categorical transition from the domain of mathematical possibility to the domain of physical fact. Once this boundary is drawn, the Bohr-Einstein debate is not resolved by declaring a winner. It is dissolved by showing that both positions describe real features of physical reality, each applicable to its proper domain.

## 2.4 The Proliferation of Interpretations: Symptoms of an Undiagnosed Condition

The history of quantum interpretation from 1927 to the present is, among other things, a history of increasingly elaborate attempts to resolve a problem that was never correctly diagnosed. The Copenhagen interpretation, the Many Worlds interpretation, the Pilot Wave interpretation, Consistent Histories, Relational Quantum Mechanics, QBism, Objective Collapse theories—each of these frameworks represents a serious attempt by serious physicists to make sense of the quantum formalism and its relationship to physical reality. Each of them contains genuine insights. And each of them, in different ways, is a response to the same unresolved foundational confusion: what does it mean for a quantum probability distribution to become a definite physical outcome?

The Many Worlds interpretation deserves particular examination here, because it represents the most dramatic consequence of failing to draw the boundary between the two domains. Proposed by Hugh Everett III in 1957, the Many Worlds interpretation begins with a simple and, within its own terms, entirely consistent observation: the Schrödinger equation is linear and deterministic, and if it is applied universally—if there is no collapse, no Process 1 in von Neumann's sense, no transition from quantum superposition to definite classical outcome—then what happens at measurement is that the measuring apparatus becomes entangled with the measured system, and the combined system evolves into a superposition of all possible measurement outcomes, each with its associated probability amplitude [12,24].

In the Many Worlds interpretation, this superposition does not collapse into one definite outcome. All outcomes occur. Each outcome exists in a separate branch of the universal wave function. The appearance of a single definite outcome—the fact that any given observer always sees one result, not many simultaneously—is explained by the fact that the observer's own physical state becomes correlated with one particular branch of the superposition through the process of measurement. The observer does not see all outcomes because the observer is themselves part of the quantum superposition, and each branch of the superposition contains an observer who sees only the outcome associated with that branch.

The Many Worlds interpretation is mathematically consistent and makes no predictions that differ from standard quantum mechanics, because it contains the same probability structure encoded in the Schrödinger equation. It also, from the perspective of this paper, represents the ultimate elaboration of the foundational confusion it inherited. The Many Worlds interpretation takes the pre-collapse quantum description domain—the domain of unrealised mathematical possibilities—and elevates every element of it to the status of physical fact. Every branch of the superposition is not a possible outcome that might be realised; it is an actually realised outcome in an actually existing branch of physical reality. The probability distribution does not describe possible physical worlds; it describes actual physical worlds, all of them equally real, distinguished only by the fact that any given observer has access to only one of them [25,26].

This is the conflation of mathematical possibility with physical reality carried to its logical extreme. The Many Worlds interpretation does not merely fail to distinguish between the pre-collapse and post-collapse domains—it abolishes the distinction entirely by declaring that everything that exists in the pre-collapse domain also exists in the post-collapse domain, only in separate branches. The wave function is not a description of possible realities; it is a description of actual realities. Probability is not a feature of our epistemic relationship to a physical world that will be one way or another; it is a feature of the distribution of observers across branches of a physical reality that is all ways simultaneously.

The intellectual cost of this position is enormous and rarely acknowledged. A physical theory that posits the continuous proliferation of entire universes—each as fully real as the one any given observer inhabits, each containing complete copies of every physical system and every observer that existed before the branching event—is a theory that has abandoned any meaningful concept of ontological

economy. It has done so in order to preserve the universality of the Schrödinger equation—to avoid the need for a collapse process that the Schrödinger equation itself cannot explain. And it has done so because the foundational confusion that treats the pre-collapse probability domain as a description of physical reality makes the Schrödinger equation's output—the full superposition of all possible outcomes—into a description of what physically exists.

Once the boundary between the two domains is correctly drawn, the Many Worlds interpretation loses its motivation. The Schrödinger equation governs the pre-collapse quantum description domain. Its output—the superposition of all possible outcomes with their associated probability amplitudes—is a description of mathematical possibilities, not physical realities. When collapse occurs, one of those possibilities becomes a physical fact. The others do not become physical facts in other universes; they simply remain what they always were—unrealised possibilities, described with mathematical precision, never instantiated as physical events. There is no need for the universe to multiply itself to accommodate them, because they were never candidates for physical existence in the first place. They were always, and only, elements of the mathematical description domain.

## 2.5 The Quantum Gravity Category Error

The relationship between quantum mechanics and general relativity is the deepest unsolved problem in theoretical physics. The two theories are each confirmed to extraordinary precision within their respective domains, and they are fundamentally incompatible with each other. The search for a quantum theory of gravity—a single theoretical framework that encompasses both—has occupied some of the finest minds in physics for nearly a century, without producing a confirmed, empirically testable result.

This paper does not claim to solve the problem of quantum gravity. It claims something more limited but more fundamental: that a significant part of the difficulty in constructing a quantum theory of gravity arises from the foundational category error that this paper identifies—the failure to distinguish between the quantum description domain and the physical event domain—and that recognising this error clarifies the nature of the problem in a way that has important consequences for how it should be approached.

General relativity is a theory of spacetime geometry. Its central mathematical object is the metric tensor, which encodes the curvature of spacetime at every point. The Einstein field equations relate this curvature to the energy-momentum tensor—a mathematical object that describes the distribution of matter and energy throughout spacetime. The energy-momentum tensor is a classical object: it assigns definite values of energy, momentum, and stress to every point of spacetime. General relativity requires, as an input, a definite physical distribution of matter and energy. It produces, as an output, the spacetime geometry determined by that distribution.

Quantum mechanics, in its pre-collapse regime, does not provide a definite physical distribution of matter and energy. It provides a probability distribution over possible physical distributions of matter and energy. The wave function of a quantum system assigns probability amplitudes to all the possible configurations the system might be found in upon measurement. Before measurement—before collapse—there is no single definite configuration. There is a superposition of all possible configurations, each weighted by its probability amplitude.

The problem of quantum gravity, at its most basic level, is this: what does the Einstein field equation look like when its source term—the energy-momentum tensor—is quantum mechanical, and therefore described by a superposition of possible configurations rather than a single definite configuration? How does spacetime respond to a quantum superposition of matter distributions?

This problem has been approached in many ways. One approach, semi-classical gravity, replaces the energy-momentum tensor with its quantum expectation value—a kind of weighted average over all possible configurations, weighted by their quantum probabilities. This approach is tractable

mathematically but known to be inconsistent: it produces results that violate fundamental physical principles in certain regimes. Another approach, canonical quantum gravity, attempts to apply quantum mechanical operators directly to the metric tensor of general relativity, treating spacetime geometry itself as a quantum variable subject to superposition and uncertainty. This approach faces severe technical difficulties and has not produced empirically confirmed predictions. String theory, loop quantum gravity, and other approaches each attempt to construct a unified framework from different starting points, with varying degrees of mathematical success and uniform absence of empirical confirmation [17,18,19,20].

The foundational difficulty underlying all of these approaches—the difficulty that none of them has yet resolved—is that they attempt to apply general relativity, a theory of definite physical configurations of matter and energy in spacetime, to a domain described by quantum mechanics—a domain of probability distributions over possible but not yet physically realised configurations. This is not a technical challenge that can be overcome by more sophisticated mathematical tools. It is a category error: the demand that a theory of physical facts accommodate descriptions of mathematical possibilities.

General relativity governs the post-collapse physical domain: the domain of definite events, definite matter distributions, definite spacetime geometries. This is what the Einstein field equations describe. They do not describe, and cannot describe, the quantum description domain—the domain of unrealised possibilities—because that domain does not contain definite matter and energy distributions that could serve as the source terms of the field equations.

A quantum theory of gravity, properly understood, is not a theory in which spacetime geometry is subject to quantum uncertainty in the same sense that particle positions are subject to quantum uncertainty. It is a theory that describes how the definite physical events produced by quantum collapse—events that occur in spacetime, that have definite energy and momentum, that curve spacetime in the way that all physical events curve spacetime—interact with and produce spacetime geometry. The quantum domain and the gravitational domain do not need to be unified into a single formalism that applies uniformly to both. They need to be related across the boundary that separates them—the collapse boundary—in a way that respects the categorical distinction between the domain of mathematical possibility and the domain of physical fact.

This reframing does not immediately produce a complete quantum theory of gravity. It does, however, identify why the standard approaches to the problem have not succeeded: they attempt a unification that presupposes the domains to be continuous when they are in fact separated by a hard categorical boundary. The correct question is not "how do we quantise gravity?" but rather "how does the collapse boundary interface with the spacetime geometry that general relativity describes?" These are different questions, and the second one is answerable in principle in a way that the first, as currently framed, is not.

The failure to ask the correct question is a direct consequence of the failure to draw the boundary. And the failure to draw the boundary is, in turn, a direct consequence of the conflation of mathematical possibility with physical reality that began—not with malice or carelessness, but with the entirely understandable haste of revolutionary discovery—in Brussels, in October 1927, when the greatest physicists in the world gathered to discuss a theory they had built but not yet understood, and left without answering the question that Lorentz had the presence of mind to ask at the very beginning: must one necessarily elevate indeterminism to a principle?

The answer is no. Indeterminism is real—within its proper domain. And its proper domain ends at collapse.

## Chapter Three: After the Boundary—The Correct Ontological Map of Quantum Mechanics

### 3.1 What Quantum Mechanics Actually Describes

Einstein wrote to Schrödinger in 1935: "Die eigentliche Schwierigkeit liegt darin, daß die Physik eine Art Metaphysik ist; Physik beschreibt Wirklichkeit. Aber wir wissen nicht, was Wirklichkeit ist; wir kennen sie nur durch die physikalische Beschreibung." Physics describes reality. But we do not know what reality is; we only know it through the physical description [21].

This remark is extraordinary in its honesty and its precision. Einstein was not claiming that physics is merely subjective, or that reality is unknowable, or that physical theories are nothing more than instruments for predicting experimental outcomes. He was identifying a genuine epistemological trap: if physical description is the only access we have to reality, then the boundary between the description and what it describes becomes invisible. This is precisely the trap that quantum foundations fell into—and that this paper argues must be made visible. Einstein himself could not escape it: believing that description is our only route to reality, he could not step outside the descriptive domain to ask whether the quantum description describes possibility or fact. That question is what this paper answers. The difficulty he identified is not a deficiency to be overcome. It is a permanent feature of the relationship between physical theory and physical reality, and any honest account of what quantum mechanics describes must begin by acknowledging it.

With this acknowledgement in place, the question of what quantum mechanics describes can be answered with a clarity that has eluded the interpretation debates of the past century. Quantum mechanics describes two things, corresponding to its two structural domains, and the relationship between those two things is mediated by the collapse boundary that separates the domains.

In the pre-collapse domain, quantum mechanics describes the complete probability structure of a physical system that has not yet produced a definite macroscopic outcome. This probability structure is not a description of our ignorance—it is not the case that the system secretly has definite properties that the wave function fails to capture. The probability structure is the complete description of the physical situation of an unrealised possibility. The wave function of an electron in a double-slit experiment does not describe an electron that is secretly going through one slit or the other; it describes a physical situation in which the question of which slit the electron goes through has no definite answer, because no definite answer yet exists. The interference pattern that results from this experiment is direct empirical evidence that the pre-collapse physical situation is genuinely probabilistic—not a reflection of our ignorance of a hidden definite trajectory, but a feature of what the physical situation actually is before collapse resolves it.

The mathematical reality of this probability structure is not in question. Wave functions are real mathematical objects. The Schrödinger equation that governs their evolution is a real physical law. The probability amplitudes that the wave function assigns to possible outcomes are real features of the physical situation they describe. What the probability structure is not is a physical distribution of matter and energy in space. It is a description of possible physical distributions of matter and energy, none of which has yet been realised as a physical fact. The distinction between a mathematical description of possible physical configurations and an actual physical configuration is the distinction between the two

domains—and it is a distinction that quantum mechanics itself, examined carefully, has always contained.

In the post-collapse domain, quantum mechanics—or rather, the physical theory appropriate to this domain—describes definite physical events: the electron detected at position A, the photon absorbed by the detector at time T, the radioactive atom that decayed in the interval between noon and one o'clock. These events are physical facts. They exist in spacetime with definite properties. They are subject to classical mechanics, classical electrodynamics, and general relativity in the appropriate regimes. They are not subject to quantum superposition, because the quantum superposition that preceded them has been resolved by collapse into the single definite outcome that each of these events represents.

The relationship between the two domains—the mechanism by which mathematical possibility becomes physical fact—is collapse. Collapse is not a dynamical process within quantum mechanics, in the sense that the Schrödinger equation is a dynamical process within quantum mechanics. It is the boundary event between the two domains: the moment at which the quantum description of an unrealised possibility terminates and a physical event comes into existence. What determines when and how collapse occurs remains one of the deepest open questions in physics, and this paper does not claim to resolve it. What this paper claims is that collapse, whatever its ultimate physical mechanism, must be understood as a transition between domains rather than a process within a domain—and that understanding it in this way clarifies, rather than deepens, the theoretical difficulties associated with it.

This is what quantum mechanics actually describes: the complete and exact mathematical structure of unrealised physical possibility, up to the moment of collapse; and the statistical distribution of definite physical outcomes, across many instances of collapse, as confirmed by the Born rule. It does not describe hidden definite properties of physical systems. It does not describe simultaneously existing branches of physical reality. It does not describe the geometry of spacetime, or the causal structure of definite physical events, or any of the other features of the post-collapse physical domain that belong to classical and relativistic physics. It describes possibility, with complete precision, up to the boundary. Everything beyond the boundary belongs to a different description domain, governed by different physical laws.

There is a sharper conclusion to be drawn here, and it has not been stated with sufficient precision in the century of debate that followed 1927. The Heisenberg uncertainty principle does not merely cease to apply after collapse. Its subject matter—the state of genuine physical indeterminacy that the principle governs—does not appear in physical reality as indeterminacy, and never will. What physical reality presents, always and without exception, is the sequence of definite facts that collapse produces: A, then B, then C, then D. The indeterminate process between A and B, the probability distribution that quantum mechanics describes with complete mathematical precision during that interval, does not appear in physical reality as indeterminacy. It appears in physical reality only as its resolution—as B, the definite fact that collapse selects. The uncertainty was real as a mathematical description of unrealised possibility. No instrument records uncertainty itself. Every instrument records an outcome.

This means that the Heisenberg uncertainty principle, while mathematically valid and physically indispensable as a description of the pre-collapse domain, has no direct instantiation in physical reality. Physical reality contains only outcomes. The uncertainty principle governs the mathematical structure of the space of possible outcomes before any outcome is selected. That structure is real—it is what makes quantum mechanics the theory it is, and it is confirmed by the statistical distribution of outcomes across many experiments. But the indeterminacy itself, as a physical state, is never what the world presents. The world presents B. The indeterminacy between A and B existed in the description. It was resolved, not observed.

### 3.2 The Dissolution of the Bohr-Einstein Debate

The Bohr-Einstein debate, as it has been understood for nearly a century, presents itself as an unresolved conflict between two irreconcilable positions on the nature of physical reality and the scope of quantum mechanics. Bohr maintained that the wave function is the complete description of a quantum system, that demanding further specification of the system's properties prior to measurement is physically meaningless, and that the uncertainty principle reflects not our ignorance but the genuine structure of quantum reality. Einstein maintained that a complete physical theory must assign definite properties to physical systems independently of measurement, that quantum mechanics fails to do this, and that therefore quantum mechanics is incomplete—a successful but provisional framework awaiting a more complete successor.

Both positions have been defended by serious physicists across the decades since 1927. The Bell theorem experiments of the 1970s and 1980s, particularly those of Alain Aspect and his collaborators, confirmed that certain classes of hidden variable theories—those that assume local definite properties for quantum systems prior to measurement—are inconsistent with the observed correlations in entangled quantum systems. These experiments are widely interpreted as vindications of Bohr's position and refutations of Einstein's. The interpretation is not incorrect, but it is incomplete. What the Bell experiments demonstrated is that the specific form of Einstein's realism—local hidden variables, definite pre-measurement properties that are independent of and prior to the measurement context—is empirically excluded. They did not demonstrate that physical reality, in the post-collapse domain, lacks the definite structure that Einstein attributed to it. They demonstrated that the pre-measurement quantum description cannot be completed by local hidden variables of the type originally envisioned by Einstein. This is entirely consistent with the analysis presented in this paper [10,27,28,29].

Once the boundary between the two domains is correctly drawn, the apparent conflict between Bohr and Einstein dissolves into a complementarity that neither man articulated but both partially perceived. Bohr's insistence that the wave function is the complete description of the quantum system is correct—within the pre-collapse domain. The probability structure of the wave function is not an incomplete description of hidden definite properties. It is the complete description of the physical situation of an unrealised possibility. There are no hidden properties beneath the wave function. The probability is all there is, prior to collapse. Bohr is right.

Einstein's insistence that physical events have definite properties independently of observation is equally correct—within the post-collapse domain. Once collapse has occurred and a physical event has come into existence, that event has definite properties. The electron detected at position A is at position A. Its position does not depend on whether anyone is looking. The moon is where it is. The physical event is a fact, and facts are not constituted by observation. Einstein is right.

The error shared by both men was the assumption that these two positions are in conflict—that one of them must be wrong. They are not in conflict. They describe different domains. Bohr's completeness claim applies to the pre-collapse quantum description domain. Einstein's realism applies to the post-collapse physical event domain. Both are correct within their proper scope. The debate between them was conducted across a boundary that neither man explicitly drew—and therefore across a gap that seemed, from both sides, to be a contradiction rather than a demarcation.

The implications of this dissolution extend beyond the historical dispute between two physicists. The Bohr-Einstein debate has shaped the intellectual culture of quantum foundations research for nearly a century. The terms of the debate—realism versus anti-realism, completeness versus incompleteness, determinism versus indeterminism—have defined the questions that quantum foundations researchers ask and the frameworks within which they seek answers. If the debate is dissolved rather than resolved—if both positions are correct about their respective domains, and the apparent contradiction between them is a consequence of failing to identify the boundary—then the questions it generated are

also transformed. The question is no longer which of the two positions is correct. The question is how to articulate the relationship between the two domains accurately, and what the implications of that relationship are for the structure of physical theory.

### **3.3 The Theoretical Landscape After the Boundary Is Drawn**

Drawing the boundary explicitly—identifying collapse as the hard ontological demarcation between the quantum description domain and the physical event domain—does not solve every open problem in quantum foundations. It does, however, clarify the structure of those problems and, in several important cases, dissolves problems that were generated by the failure to draw it.

The measurement problem, as classically formulated, is the problem of explaining how a linear, deterministic quantum evolution produces non-linear, probabilistic measurement outcomes. Within the framework of this paper, this problem is reconceived. The Schrödinger equation governs the pre-collapse domain. Collapse is the boundary between the domains. The question of how the Schrödinger equation produces collapse is analogous to the question of how a map produces the territory it depicts—it is a category error to expect the governing equation of one domain to explain the transition to another. What can be asked, and what remains a genuine open question, is what physical conditions determine when collapse occurs—what physical process constitutes the boundary event itself. This is a real question, but it is a different question from the measurement problem as classically stated, and it does not require a solution that preserves the universality of the Schrödinger equation across both domains.

Decoherence theory, developed from the 1970s onward by H. Dieter Zeh, Wojciech Zurek, and others, has provided a detailed account of how quantum superpositions become effectively classical through interaction with the environment. When a quantum system interacts with a large environment—an apparatus, an air molecule, a photon—the quantum coherence between different components of the superposition is distributed into the enormous Hilbert space of the combined system and environment. The interference terms that characterise quantum superposition become practically unmeasurable—suppressed to levels far below any experimental sensitivity. Decoherence is real and important, and it explains why macroscopic objects do not exhibit quantum superposition effects despite being composed of quantum mechanical constituents [14,30,31].

Within the framework of this paper, decoherence describes a process within the pre-collapse domain: the progressive entanglement of a quantum system with its environment, leading to the effective disappearance of quantum coherence between possible outcomes. Decoherence does not, by itself, produce collapse—it does not select one outcome as the actual outcome, eliminating the others as unrealised possibilities. It explains why the quantum superposition becomes effectively invisible at the macroscopic scale, but it does not explain why the world presents one definite macroscopic fact rather than a superposition of facts. The gap between decoherence and definite outcome is precisely the collapse boundary. Decoherence describes the approach to the boundary from within the pre-collapse domain; it does not describe the crossing of the boundary or what lies beyond it [15,32].

The correct relationship between decoherence and collapse, within this framework, is that decoherence identifies the physical conditions under which collapse becomes effectively complete—the conditions under which the quantum superposition has become so thoroughly entangled with its environment that no practical measurement could detect the coherence between its branches. In macroscopic systems interacting with thermal environments, this occurs on timescales that are astronomically small compared to any observable dynamical timescale. This is why macroscopic physical events appear definitively classical: not because quantum mechanics does not apply to them, but because decoherence drives them to the collapse boundary so rapidly that the pre-collapse quantum description domain is effectively inaccessible at the macroscopic scale.

For the quantum gravity problem, drawing the boundary suggests a reorientation of the research programme that, while not immediately yielding a complete theory, identifies the correct structural relationship between the two frameworks. General relativity describes the geometry of spacetime as produced by the distribution of matter and energy in the post-collapse physical event domain. Quantum mechanics describes the probability structure of physical systems in the pre-collapse description domain. These two theories do not need to be unified into a single formalism that applies uniformly to both domains, because the domains are categorically distinct. What is needed is an account of how physical events in the post-collapse domain—events that have definite energy and momentum, that curve spacetime in the way all physical events curve spacetime—arise from the collapse of quantum probability distributions, and how the spacetime geometry produced by those events feeds back into the boundary conditions for subsequent quantum evolution.

This is a genuinely difficult problem, and solving it in detail lies beyond the scope of this paper. But framing it correctly is not trivial. The currently dominant approaches to quantum gravity—string theory, loop quantum gravity, canonical quantum gravity—all attempt to construct a unified framework in which spacetime geometry is itself subject to quantum uncertainty, treated as a quantum variable that can exist in superposition. Within the framework of this paper, this approach conflates the two domains: it attempts to bring the post-collapse description of spacetime geometry into the pre-collapse quantum description domain, treating a feature of the physical event domain as though it were a quantum probability distribution. The difficulties these approaches face—the non-renormalisability of gravity in perturbative treatments, the problem of time in canonical quantum gravity, the landscape problem in string theory—are at least partly consequences of this conflation [17,33].

The correct question for quantum gravity is not how to quantise the gravitational field in the same sense that electromagnetic and other fields have been quantised—treating spacetime geometry as a quantum variable subject to superposition. The correct question is how the collapse boundary interfaces with spacetime geometry: how definite physical events, produced by collapse, generate the spacetime curvature that general relativity describes, and how that curvature determines the background against which subsequent quantum evolution occurs. This is a relational question about two distinct domains and the boundary between them, not a unification question about a single domain governed by a single formalism [18,19,20].

The Many Worlds interpretation, examined within this framework, is revealed as a theoretical structure built entirely on the refusal to draw the boundary. By postulating that the Schrödinger equation governs reality universally and completely, with no collapse and no transition between domains, the Many Worlds interpretation converts the entire pre-collapse description domain into physical reality. Every branch of the universal wave function is a real physical world. The probability distribution that quantum mechanics assigns to possible outcomes becomes a distribution over actual worlds, each as real as any other. This move achieves consistency within its own terms—there is no measurement problem, because there is no collapse, and the apparent definiteness of any individual observer's experience is explained by the observer's entanglement with one particular branch of the superposition. But the price paid is the abandonment of any distinction between mathematical description and physical reality. The wave function is not a description of what might happen. It is a description of what does happen—everywhere, in every branch, simultaneously and permanently. Possibility and actuality are identified, and the physical world proliferates without bound to accommodate every quantum probability amplitude as an actual physical fact [12,24,25,26]

Once the collapse boundary is drawn, this proliferation is unnecessary. The wave function describes what might happen, with complete mathematical precision. Collapse selects what does happen. The branches that are not selected do not become real in other universes; they remain what they always were—elements of the mathematical description of unrealised possibility. No universe is created or

destroyed by this account. The formalism is preserved. The boundary is respected. And the ontological economy that any serious physical theory should aspire to is maintained.

### 3.4 A Completed Answer to Lorentz's Question

Lorentz asked, in October 1927, whether one must necessarily elevate indeterminism to a principle. The question remained unanswered for the duration of the Fifth Solvay Conference, and it has remained unanswered, in formal terms, ever since. The framework developed in this paper provides the answer [1].

Indeterminism is real. It is not an artefact of our ignorance, not a consequence of insufficient measurement precision, not a placeholder for hidden variables that a future theory will reveal. The probability structure of the pre-collapse quantum description domain is genuine: there is no hidden fact about which slit the electron goes through, no hidden fact about where the radioactive atom will decay, no hidden fact about the spin of the particle before its spin is measured. The indeterminism is constitutive of the physical situation of an unrealised possibility. To this extent, the answer to Lorentz's question is yes: within the pre-collapse domain, indeterminism must be elevated to a principle, because the physical situation it describes is genuinely indeterminate.

But this yes is bounded. It applies within a domain that is itself bounded—the domain of unrealised physical possibility, governed by quantum mechanics, terminated at the collapse boundary. Beyond that boundary lies the domain of realised physical fact, governed by classical and relativistic physics, in which determinism—or at any rate, definiteness—is restored. The electron detected at position A is at position A. The radioactive atom that decayed between noon and one o'clock decayed in that interval. The physical event is definite. Indeterminism does not follow the event across the collapse boundary. It remains behind, in the domain where it belongs—the domain of what might have happened, described with mathematical completeness by the wave function that has now been resolved into fact.

Einstein was right that results are results. He was right that facts are facts. He was right that the moon is where it is regardless of whether anyone measures it. He was right about everything in the post-collapse domain. And Bohr was right that the pre-collapse quantum description cannot be supplemented by hidden definite properties without contradicting the experimental record. Both men were right. The debate between them was a debate conducted without a map—without the explicit identification of the boundary that separates the territory each man was correctly describing.

The century of confusion that followed from that missing map has been, in its way, extraordinarily productive. The interpretation debates generated by the failure to draw the boundary produced decoherence theory, quantum information theory, the Bell inequalities and their experimental tests, and a rich landscape of theoretical frameworks that have illuminated, from many different angles, the structure of quantum mechanics and its relationship to physical reality. None of this work is wasted. All of it contributes to the understanding of the two domains and the boundary between them.

What has been missing is the map itself: the explicit identification of collapse as the hard ontological boundary between the domain of mathematical possibility and the domain of physical fact; the recognition that Heisenberg's uncertainty principle governs the pre-collapse domain and terminates at the collapse boundary; the understanding that quantum probability distributions are not physical distributions of matter and energy and cannot serve as the source terms for the Einstein field equations; and the dissolution of the Bohr-Einstein debate through the recognition that both positions are correct about their respective domains.

This paper advances the following conclusions. The domain of quantum mechanical description is the domain of objective probability structures over possible measurement outcomes. The wavefunction  $|\psi\rangle$  is a mathematically well-defined entity that encodes these possibilities; it does not represent a

simultaneously existing set of classical physical facts. Physical actuality belongs to the domain of realised outcomes. No definite physical property may be attributed to a quantum system independently of the physical conditions under which that property becomes actualised. These conclusions are not interpretive additions to quantum mechanics: they are what the formalism itself requires.

Bohr's foundational delimitation—that quantum mechanics does not describe a pre-existing classical reality independently of the conditions under which physical phenomena are defined [9,34]—was never formalised as a structural constraint, remaining an interpretive principle rather than an explicit boundary condition. This paper therefore holds that Bohr's foundational delimitation is the explicit definition and boundary of quantum mechanics. It is the only foundational reading of quantum mechanics, the one from which the formalism cannot be separated without logical drift. Any departure does not produce an alternative interpretation; it produces a distortion, as the analyses of many-worlds, parallel branches, and wavefunction realism throughout this paper demonstrate. The staged mapping  $|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$  [35] formalises this boundary, specifying the structural interface between the domain of objective probability structures and the domain of physically actualised facts. Bohr's delimitation is not a philosophical preference. It is the boundary the formalism itself draws.

Physics describes reality. The description has two parts, corresponding to two domains, separated by a boundary. The boundary is collapse. Once this is clearly seen, the question Lorentz asked in 1927 can finally be answered: indeterminism is a principle—within its domain. And its domain has an edge.

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